

Theory of ice premelting in porous media

Hendrik Hansen-Goos^{1,*} and J. S. Wettlaufer^{1,2,3}

¹*Department of Geology and Geophysics, Yale University, New Haven, CT 06520, USA*

²*Department of Physics, Yale University, New Haven, CT 06520, USA*

³*Program in Applied Mathematics, Yale University, New Haven, CT 06520, USA*

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Premelting describes the confluence of phenomena that are responsible for the stable existence of the liquid phase of matter in the solid region of its bulk phase diagram. Here we develop a theoretical description of the premelting of water ice contained in a porous matrix, made of a material with a melting temperature substantially larger than ice itself, to predict the amount of liquid water in the matrix at temperatures below its bulk freezing point. Our theory combines the interfacial premelting of ice in contact with the matrix, grain boundary melting in the ice, and impurity and curvature induced premelting, the latter occurring in regions which force the ice-liquid interface into a high curvature configuration. These regions are typically found at points where the matrix surface is concave, along contact lines of a grain boundary with the matrix, and in liquid veins. Both interfacial premelting and curvature induced premelting depend on the concentration of impurities in the liquid, which, due to the small segregation coefficient of impurities in ice are treated as homogeneously distributed in the premelted liquid. Our principal result is an equation for the fraction of liquid in the porous medium as a function of the undercooling, which embodies the combined effects of interfacial premelting, curvature induced premelting, and impurities. The result is analyzed in detail and applied to a range of experimentally relevant settings.

I. INTRODUCTION

Due to advances in our understanding of the premelting of ice and other materials over the last fifteen years [1], and an increasing appreciation of the influence and extent of this phenomenon in technological, terrestrial and extraterrestrial settings, it is prudent to revisit the detailed thermodynamic basis of its manifestation in porous media where water is invariably found in adsorbed layers or in bulk. In 1992 Cahn, Dash, and Fu (CDF) [2] developed the original premelting based theory for the liquid fraction f_l of water in a porous medium at sub-freezing temperatures. Given experimental and theoretical advances in the interim, we extend their work in a number of respects. CDF considered a matrix of monodisperse spheres arranged either in a simple cubic packing or in a cubic close packing, regarded as lower and upper bounds for the experimental systems. We design our theory for the *random* close packing of monodisperse spheres, which may be a more realistic description for most applications, e.g. frozen soils, where even a small degree of polydispersity prevents a periodic packing of the matrix particles. The assumed periodic packing of the matrix particles leads CDF to take the grain size of the interstitial ice to be on the order of the size of the matrix lattice unit cell. Given that the degree of polycrystallinity of the interstitial ice has a complicated dependence on the freezing rate [see e.g., Ref. 3, and references therein] and the sample annealing history, the ice grain size may differ substantially from the matrix particle size. Whence, polycrystallinity of the interstitial ice

enters into our theory as a parameter, namely the density of grain boundary surface relative to the interstitial volume. Calculations since CDF showed that the dispersion force contribution to interfacial melting could be repulsive or attractive [1, 4] and thus favor or suppress melting in a manner and strength that depends on the effective dielectric response of the ice/water/matrix interface. Moreover, it has been understood that the great inter-experimental variability in both the interfacial and surface melting of ice was likely due to the sensitivity of the phenomena to impurities which have a rich and complex influence on the melting behavior [5]. Thus, our theory allows for both attractive and repulsive dispersion forces of arbitrary strength (depending on the matrix material) and includes the effect of charges at the interface, which give rise to an electrostatic force. Finally, most importantly, our work includes the effect of charged impurities in the premelted liquid on both interfacial and curvature induced premelting.

The article is structured as follows. In Section II we lay out the tenets and details of our theory, discussing interfacial premelting and curvature induced premelting in the presence of charged impurities and combine them in order to describe premelting in a porous medium. The final result of the theoretical treatment is an implicit functional equation for the liquid fraction f_l , Eq. (21). The first part of Section III is dedicated to the analysis of interfacial premelting as described by Eq. (7), and in the second part Eq. (21) for the liquid fraction f_l is analyzed by extracting exact solutions for certain limiting cases and numerical solution in the general case. The porous media which we consider consist of fused quartz, gold, and silicon. We summarize in Section IV.

*Electronic address: hendrik.hansen-goos@yale.edu